

Structure of Amine:

- The hybridisation of 'N' atom in trimethyl amine is, **(E¹-25)** a) sp^3 b) sp^2 c) sp d) dsp^2
- The IUPAC name of $CH_3-NH-CH_2-CH_3$ is
a) N-methyl ethanamine b) Methyl ethanamine
c) N,N-methyl ethanamine d) N-ethyl ethanamine

- Write IUPAC name of the following $CH_3-CH_2-N-(CH_3)_2$

Preparation of Amines

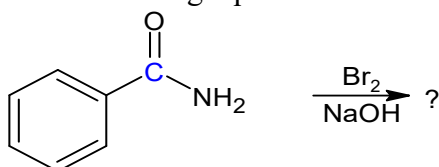
- Explain the reduction of nitro compounds to amines with an example **(S-16)** OR How does nitrobenzene is reduced to aniline? **(A-18)** Give equation OR How is aniline prepared from nitrobenzene? **(A-20,22)**
- Identify the reactant 'A' in the following reaction : $A + 2R-X \rightarrow R_4N^+X^-$ **(A-15)**
- The final product obtained in ammonolysis of alkyl halides is **(E²-25)**
a) quaternary ammonium salt b) primary amine c) secondary amine d) tertiary amine
- Amides on reduction with lithium aluminium hydride yield
a) Nitriles b) Amines c) Alcohols d) Aldehydes

Gabriel phthalimide synthesis:

- Explain the preparation of primary amine by Gabriel phthalimide synthesis . Write the equation
- Write the chemical equations involved in the Gabriel phthalimide synthesis of methanamine **(E¹-24)**
- Why cannot aromatic primary amines be prepared by Gabriel phthalimide synthesis? **(S-16)**
- Which of the following amines cannot be prepared by Gabriel Synthesis.
a) methanamine b) Ethanamine c) Propanamine d) Aniline
- Gabriel phthalimide synthesis is used to prepare ___ amines only

Hoffman bromamide synthesis

- Explain Hoffmann bromamide degradation reaction with example **(S-15,16,21)** OR How primary amine is prepared by Hoffmann bromamide degradation reaction? write the equation **(A-17)**
- How is methanamine prepared by Hoffmann bromamide degradation reaction? Give equation **(S-18)**
OR Explain Hoffmann's bromamide degradation reaction for the preparation of methanamine. **(A-15)**
- Complete the following: $CH_3CONH_2 + Br_2 + 4 NaOH \rightarrow ?$ **(S-14, A-15,19)** OR
 $R-CO-NH_2 \xrightarrow{Br_2 / KOH} ?$
- Explain Hoffmann bromamide degradation for the preparation of aniline **(A-14)** OR How is aniline is prepared by Hoffmann bromamide degradation reaction ? Give equation **(S-19)**
- Complete the following equation



18. The amide which gives ethanamine by Hoffmann bromamide reaction is
 a) methanamide b) Ethanamide c) Propanamide d) butanamide
19. Hofmann's Bromamide reaction is to convert
 a) Acid to alcohol b) Alcohol to acid c) Amide to amine d) Amine to amide
20. The product obtained when propionamide subjected for Hoffmann degradation
 a) Methyl amine b) Propyl amine c) Ethyl amine d) Ethyl bromide
21. Why Hofmann's bromamide reaction is called degradation reaction?
- Ans: Product has one carbon lesser than reactant (amide)**
22. How do you convert an amide into primary amine having one carbon atom less than the starting compound? Name the reaction.
23. Amide react with bromine and alkali to form primary amine. The reaction is known as
 a) Hoffmann reaction b) Kolbe's reaction c) Etard reaction d) Cannizzaro reaction

Physical properties:

Physical Properties (ಭೌತಿಕ ಗುಣಲಕ್ಷಣಗಳು)

Amines	Soluble in water	Reason
Lower aliphatic amines	Soluble	form hydrogen bonds with water molecules
Higher aliphatic amines	Less soluble	increase in size of the hydrophobic alkyl part.
Boiling point of primary and secondary amines more than tertiary amines : have stronger intermolecular to hydrogen bonding		

$pK_b = -\log K_b$		
K_b	pK_b	
larger	small	Stronger base
small	large	Weaker base

Electron donating group (+I effect)	Increases basicity (Stronger base)	Bcz increase in electron density on Nitrogen atom
Electron withdrawing group (-I effect)	Decrease in basicity (weaker base)	Bcz decrease in electron density on Nitrogen atom

24. Give reason
 a) Aliphatic amines of lower molecular mass soluble in water Ans:
 b) In the isomeric amines Butanamine has more boiling point than N, N – dimethylmethanamine
 c) Tertiary amine have low boiling point than primary amines and secondary amines due to absence of _____
 d) Solubility of $C_2H_5NH_2$ in water is due to the formation of _____ bonding
25. In case of amines, the melting points are normally highest for
 a) Primary amines b) Secondary amines c) Tertiary amines d) Quaternary ammonium salts
26. Arylamines get coloured on storage due to atmospheric _____

Basicity:

27. Among aryl amines and ammonia which is more basic? (E¹-25)

28. Which is more basic among aqueous solution of aniline and ammonia?? Give one reason (A-15,) OR Give reason: aniline is a weaker base than ammonia OR Between ammonia and aniline, which is more basic? (S-18) Ans:
29. Between methanamine (CH_3NH_2) and aniline benzenamine (aniline)($\text{C}_6\text{H}_5\text{NH}_2$) which is more basic? Give reason (A-14,19)
30. Give reason : Dimethylamine is more basic than methyl amine.
31. Amongst the following amines, the strongest base in aqueous medium is
a) CH_3NH_2 b) $(\text{CH}_3)_3\text{N}$ c) $(\text{CH}_3)_2\text{NH}$ d) $\text{C}_6\text{H}_5\text{NH}_2$
32. Aniline is_____ basic than ammonia
33. Give reason: Aromatic amines are weaker bases than ammonia
34. The amine with more basicity in gaseous phase is_____
35. Between Arylamine & ammonia which has higher $\text{P}^{\text{K}}_{\text{b}}$ value
a) Arylamine higher b) Ammonia higher c) Arylamine lower d) None of these
36. Larger the $\text{p}^{\text{K}}_{\text{b}}$ values of amine, weaker is the _____
37. Which of the following has highest $\text{p}^{\text{K}}_{\text{b}}$ value?
a) $(\text{CH}_3)_3\text{CNH}_2$ b) NH_3 c) $(\text{CH}_3)_2\text{NH}$ d) CH_3NH_2
38. Arrange the following compounds, in the increasing order of their basic nature in aqueous solution. Benzamine($\text{p}^{\text{K}}_{\text{b}}=9.38$), Ethanamine($\text{p}^{\text{K}}_{\text{b}}=3.29$), Phenylmethanamine($\text{p}^{\text{K}}_{\text{b}}=4.70$)
39. Identify the more basic compound among Aniline, 4-Nitroaniline, 4-methylaniline
40. The correct order of basic character among the following in aqueous solution
i) NH_3 ii) CH_3NH_2 iii) $(\text{CH}_3)_2\text{NH}$ iv) $(\text{CH}_3)_3\text{N}$
a) i<iv<ii<iii a) iv<i<ii<iii a) iv<ii<i<iii a) i<ii<iii<iv
41. Which of the following order of basic strength of amines in aqueous solutions is correct?
a) $(\text{CH}_3)_2\text{NH} > \text{CH}_3\text{NH}_2 > (\text{CH}_3)_3\text{N}$ b) $\text{CH}_3\text{NH}_2 > (\text{CH}_3)_3\text{NH} > (\text{CH}_3)_2\text{N}$
c) $(\text{CH}_3)_2\text{N} > (\text{CH}_3)_2\text{NH} > \text{CH}_3\text{NH}_2$ d) $(\text{CH}_3)_3\text{N} > (\text{CH}_3)_2\text{NH} > \text{CH}_3\text{NH}_2$

Reaction with Nitrous acid:

42. Name the major product formed write the equation when nitrous acid is treated with methylamine (S-15,17)
43. The gas liberated when aliphatic primary amines reacts with HNO_2 is_____
44. When Methylamine reacts with nitrous acid, it produces _____ with liberation of nitrogen gas.(E²-24)
45. The gas liberated when aliphatic primary amines reacts with HNO_2 is_____
46. The final product formed when a primary aliphatic amine reacts with nitrous acid [HNO_2] is____
a) Alcohol b) Diazonium salt c) Isocyanide d) Amide
47. The gas liberated when ethyl amine reacts with HNO_2 is_____

Acylation:

48. Mention the role of pyridine in acylation of amines **Ans:**

Carbylamine reaction:

49. Explain the carbylamines reaction by taking methyl amine as an example (A-14,16,18,S-14) OR How does primary amine undergo carbylamine reaction? Give the chemical equation (S-19, E²-24)
50. What is the product formed when aniline is heated with chloroform and ethanolic potassium hydroxide? Write the equation. (S-22)
51. Carbylamine reaction is shown by (E³-24)
- a) Primary amines b) Secondary amines c) tertiary amines d) Primary alcohols
52. Which of the following gives carbylamines reaction
- a) Aniline b) N-methylaniline c) N,N-dimethylaniline d) dimethylamine
53. How would you convert methyl amine into methyl isocyanide
54. The bad smelling substance formed by the action of alc.KOH on chloroform and aniline is
- a) Nitrobenzene b) Phenyl isocyanide c) Phenyl cyanide d) Phenyl thiocyanide
55. Mention the foul smelling substances formed when a primary amine is heated with chloroform and alcoholic potash
56. Carbylamine reaction is answered by
- a) Carboxylic acid b) 3⁰ amines c) 2⁰ amines d) 1⁰ amines

Hinsberg's reagent test

Quick revision:

Reaction with **Hinsberg's reagent**

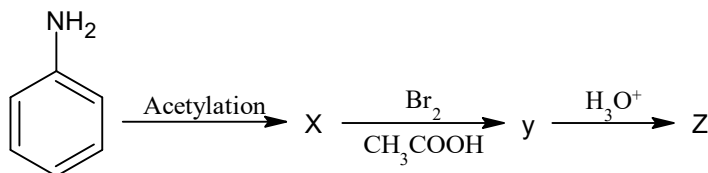
Primary amine	React	Soluble in base (alkali)	Due to presence of acidic hydrogen
Secondary amine	React	Insoluble in base	Due to absence of acidic hydrogen
Tertiary amine	Do not react		No replaceable hydrogen

57. How do you distinguish between primary amine and secondary amine using Hinsberg's reagent? (E²-25)
58. Hinsberg's reagent is (KCET-20)
- a) C₆H₅SO₂Cl b) C₆H₅SO₂NH₂ c) CH₃COCl / pyridine d) (CH₃CO)₂O / pyridine
59. What is Hinsberg reagent? (A-14)
60. Write the structure of Hinsberg reagent. Which type of amines reacts with Hinsberg reagent to give precipitate, which is soluble in NaOH? Write its chemical reaction. (E²-24)
60. 3⁰ amine doesn't answer Hinsberg test, because _____
- a) It forms salt with Hinsberg reagent b) It has no replaceable hydrogen
- c) It is not reactive d) It has replaceable hydrogen
61. Explain the reaction of ethanamine with Hinsberg's reagent. Why is the product obtained in the above reaction soluble in alkali?

- 62.** Which of the following amine does not react with Hinsberg's reagent?
- a) Primary amine b) tertiary amine
c) Secondary amine d) Both Primary and secondary amine
- 63.** If the product obtained by an amine with Hinsberg's reagent is insoluble in NaOH solution, then the amine is
- a) Primary amine b) Secondary amine c) Tertiary amine d) Either a) or c)
- 64.** Benzene sulphonyl chloride will not precipitation with_____
- 65.** Which statement is not correct for Benzene sulphonyl chloride?
- a) It is known as Hinsberg's reagent
b) It is used to distinguish primary and secondary amines.
c) On treatment with secondary amine, it leads to form a product that is soluble in alkali.
d) It doesn't react with tertiary amines.

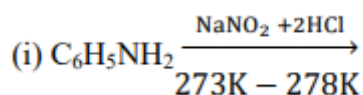
Electrophilic substitution reaction:

66. To prepare p-Nitroaniline as a major product from aniline, the amino group is protected by; (**E¹-24**)
a) Acetylation b) Alkylation c) Saponification d) Sulphonation
67. Explain with a chemical equation for the conversion of aniline to 4-bromoaniline (**S-20, E²-25**)
68. Nitration of aniline in acidic medium will yield significant amount of meta-derivative along with ortho and para derivatives. Why?
69. Aniline on reaction with excess of bromine water gives
a) p-bromo aniline b) o-bromo aniline c) 2,4,6-tri bromo aniline d) m-bromo aniline
70. Identify the products X, Y and Z in the following conversion



Benzene diazonium salt

- 71.** Explain the conversion of aniline to benzene diazonium chloride with chemical equation.(E¹-25) OR Name the major product formed write the equation when nitrous acid is treated with : Aniline at low temperature (S-15) OR How is aniline converted into benzene diazonium chloride? Give equation(S-18).
- 72.** Complete the following reactions by giving major products (A-23)



73. The reagent that can be used to convert benzene diazonium chloride to benzene is (E²-25)
a) $\text{H}_3\text{PO}_2 + \text{H}_2\text{O}$ b) SnCl_2/HCl c) LiAlH_4 d) H_3PO_4
74. The formula of benzenediazonium chloride is _____ (E¹-24)
75. How do you convert a diazonium salt solution into iodobenzene? Give equation (S-23)
76. Benzene diazonium chloride when heated with water produces (KCET-84)

- a) Phenol b) Aniline c) Benzyl alcohol d) Chlorobenzene

77. $\text{ArN}_2^+\text{X}^- \xrightarrow{\text{Cu/HCl}} \text{ArCl} + \text{N}_2 + \text{CuX}$. This reaction is referred as **(E³-24)**

- a) Sandmeyer reaction b) Gattermann reaction c) Diazotisation d) Coupling reaction

78. The Benzene Diazonium Chloride (BDC) reacts with Aniline in acidic medium at 273-278K to form coupling compound known as

- a) p-amino azobenzene b) O-amino azobenzene
c) m-amino azobenzene d) all the above

79. During diazotization, the nitrous acid is produced in the reaction mixture by the reaction of

- a) NaNO_2 & HCl b) NaNO_3 & HCl c) NaNO_3 & HNO_3 d) NaNO_3 & HNO_3

80. Choose the most stable diazonium salt

- a) $\text{C}_6\text{H}_5\text{CH}_2\text{N}_2^+\text{X}^-$ b) $\text{CH}_3\text{N}_2^+\text{X}^-$ c) $\text{C}_6\text{H}_5\text{N}_2^+\text{X}^-$ d) $\text{CH}_3\text{CHN}_2^+\text{X}^-$

81. Why diazonium salt is generally not stored and is used immediately after its preparation?

Ans: It is very unstable and dissociates to give Nitrogen

82. An aromatic compound A reacts with nitrous acid ($\text{HNO}_2 + \text{HCl}$) at 273-278K to give a product B. It reacts with water to give phenol. Name the product A and B. Write the balanced chemical equations

83. On heating Benzene diazonium chloride with hypophosphorous acid the product obtained is _____

84. Benzene diazonium chloride reacts with cuprous chloride in HCl to form chlorobenzene and _____ gas

85. The conversion of benzene diazonium salt to haloarenes in presence of copper in acidic medium is called _____

86. Benzene diazonium chloride when warmed with water gives

- a) Phenol b) Chlorobenzene c) Benzene d) Aniline

87. The reagent used to convert benzenediazonium chloride to benzene is _____

- a) H_2O b) $\text{Cu}_2\text{Cl}_2/\text{HCl}$ c) Cu/HCl d) $\text{CH}_3\text{CH}_2\text{OH}$

Coupling reactions

88. Explain the coupling reaction of Benzene diazonium chloride with phenol. Write the equation. **(S-22, E¹-24)**

89. Discuss the coupling reaction of benzene diazonium chloride with aniline **(S-23)**

90. The colour of p-aminoazobenzene is

- a) Red b) orange c) Green d) yellow

91. Reaction of aniline with benzene diazonium chloride in acidic medium gives _____

92. p-hydroxyazobenzene is a _____

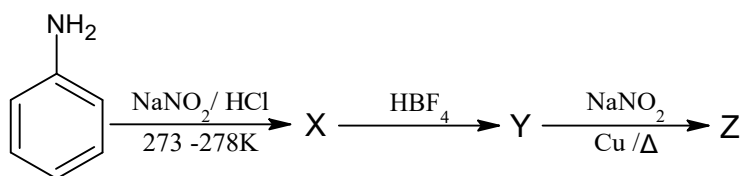
93. Benzene diazonium chloride reacts with phenol to form p-hydroxy azobenzene in

- a) acidic medium b) neutral medium c) basic medium d) both acidic and neutral medium

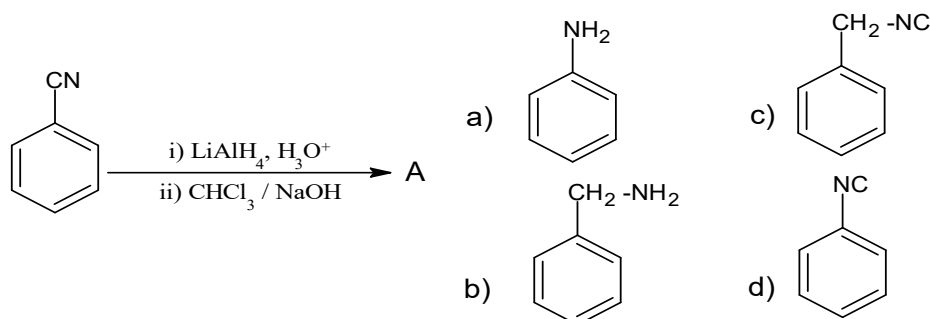
94. In the coupling reaction of $\text{C}_6\text{H}_5\text{N}_2^+\text{Cl}^-$ with $\text{C}_6\text{H}_5\text{OH}$, the coupling takes place at

- a) Meta position of $\text{C}_6\text{H}_5\text{OH}$ b) Para position of $\text{C}_6\text{H}_5\text{OH}$
c) Para position of $\text{C}_6\text{H}_5\text{N}_2^+\text{Cl}^-$ d) Meta position of $\text{C}_6\text{H}_5\text{N}_2^+\text{Cl}^-$

95. Identify X,Y,Z



96. The product formed from the following reaction sequence is



97. An organic compound **P** with molecular formula $\text{C}_6\text{H}_7\text{N}$ is treated with acetic anhydride in the presence of pyridine gives compound **Q**. The compound **Q** undergoes bromination with bromine in acetic acid gives substituted amide **R**. The compound **R** on acid hydrolysis yields a substituted amine **S**. Write all the reactions involved.

98. An aromatic compound 'A' on treatment with aqueous ammonia and heating forms a compound 'B' which on heating with Br_2 and KOH forms a compound 'C' of molecular formula $\text{C}_6\text{H}_7\text{N}$. Write the structure and IUPAC name of compounds A, B and C.

Match the columns

Column-I	Column-II
(A) $\text{ArN}_2^+\text{Cl}^- \longrightarrow \text{ArOH}$	(p) $\text{HBF}_4 / \text{NaNO}_2$
(B) $\text{ArN}_2^+\text{Cl}^- \longrightarrow \text{ArNO}_2$	(q) H_2O
(C) $\text{ArN}_2^+\text{Cl}^- \longrightarrow \text{ArH}$	(r) HBF_4
(D) $\text{ArN}_2^+\text{Cl}^- \longrightarrow \text{ArF}$	(s) $\text{CH}_3\text{CH}_2\text{OH}$

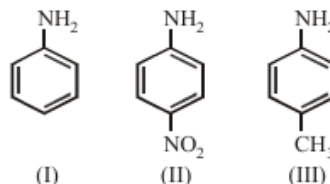
Match the columns

Column-I	Column-II
(A) Benzene sulphonyl chloride	(p) Zwitter ion
(B) Sulphanilic acid	(q) Hinsberg reagent
(C) Alkyl diazonium salts	(r) Dyes
(D) Aryl diazonium salts	(s) Conversion to alcohols

The correct increasing order of basic strength for the following compounds is

The reagents that can be used to convert benzenediazonium chloride to benzene are

- (i) SnCl_2/HCl (ii) $\text{CH}_3\text{CH}_2\text{OH}$
 (iii) H_3PO_2 (iv) LiAlH_4



Column I (Reaction)

Column II (Description/Product)

(A) Carbylamine reaction

(p) A test for the detection of primary amines

(B) Gabriel synthesis

(q) Used for the preparation of pure primary amines

(C) Hoffmann bromamide reaction

(r) Amide is converted to an amine with one less carbon

Column I (Reaction/Property)	Column II (Product/Reagent/Explanation)
(A) Carbylamine Reaction	(p) Benzene diazonium chloride
(B) Hofmann Bromamide Degradation	(q) Isocyanides (Foul smell)
(C) Diazotization	(r) RNH_2 with one less carbon atom
(D) Hinsberg's Reagent	(s) Differentiate 1° , 2° , and 3° amines
(E) $\text{C}_6\text{H}_5\text{NH}_2$ is a weaker base than CH_3NH_2	(t) Lone pair of N is delocalized over the benzene ring

Column A	Column B
A. Aniline + Br_2 water	1. Benzene diazonium chloride
B. Aniline + NaNO_2/HCl ($0-5^\circ\text{C}$)	2. 2,4,6-tribromoaniline
C. Benzene diazonium chloride + phenol	3. Azo dye
D. Aniline + acetyl chloride	4. Acetanilide